



COMPUTER SECURITY

2025

FINAL 2020 IT
(Solved)



SENIOR

25

BY AB & AE & AM & MR

Join our Telegram!
@qahwa_is



Choose the correct answer:

1.	Steganography uses _____ in graphic image to hide important data			
	a) LSB	b) HSB	c) MSB	d) NSB
2.	_____ is an attempt to discover PW by attempting to use every possible PW from a predefined list .			
	a) Brute-force attack	b) Dictionary attack	c) Sniffing attack	d) Cracking
3.	Timely response is an expectation of _____ security goal			
	a) Integrity	b) Availability	c) Confidentiality	d) All are true
4.	_____ Controls access on the basis of packet address or <u>specific transport protocol type</u>			
	a) Stateful inspection	b) Screening routers	c) Application proxies	d) Guards
5.	_____ replicates a copy of itself <u>to other systems</u> to execute a malicious code			
	a) Virus	b) Worm	c) Trojan horse	d) Trapdoor
6.	A firewall system characterized by being two-headed device is			
	a) Screening routers	b) Stateful inspection	c) Application proxies	d) Guards
7.	_____ Builds a model of acceptable behavior and flag exceptions to that model			
	a) Normal based	b) Signature-based IDS	c) Heuristic IDS	d) Honey pot
8.	It attracts the intruder , record his actions, and even attempt to trace who the attacker is from action, packet data, or connections.			
	a) Heuristic IDS	b) Honey pot	c) Normal Based	d) Signature-based IDS
9.	Intercepting and <u>carrying on a session begun by another entity</u> is known as _____ attack			
	a) Eavesdrop	b) Wiretap	c) Spoofing	d) Session hijacking
10.	_____ this form of protection limits not just the access to an object but the use made of that object after it has been access			
	a) Share via access limitation	b) Share all or share nothing	c) Share Via capabilities	d) Limit use of an object
11.	_____ is when one host pretends to be another .			
	a) Eavesdrop	b) Spoofing	c) Wiretap	d) Session hijacking
12.	Consider Case 2, as shown in table 1, the suitable firewall type is _____			
	a) Personal Firewall	b) Screening routers	c) Application Proxies	d) Guards
13.	With this form of protection, the owner of an object declares it to be public or private			
	a) Share via access limitation	b) Share all or share nothing	c) Don't protect model	d) Isolate mode
14.	Consider case 1 , as shown in table 1 , the suitable firewall type is _____			
	a) Screening routers	b) Stateful inspection	c) Physical separation	d) Application proxies
15.	_____ in which processes conceal their data and computations in such a way that they are unintelligible to outside processes			
	a) Temporal Separation	b) Logical Separation	c) Physical separation	d) Cryptographic separation



16.	DES encrypts data in <u>blocks of size</u> _____, using a <u>key of length</u> _____																															
	a)	64- Bit , 32- Bit	b)	32- Bit , 32- Bit	c)	32- Bit , 56-Bit	d)	64-Bit , 56-Bit																								
17.	_____ Change control on the OS instructions, when a computer is started																															
	a)	Memory resident virus	b)	Boot sector virus	c)	One time execution	d)	All																								
18.	Encrypting p = "computers security" using row transposition algorithm K = 425163 , then C is _____																															
	In row transposition write the plaintext in 6 columns بناء على عدد ارقام المفتاح																															
	Then rearrange the columns ascendingly رتب العواميد تصاعدي																															
	Then write column per column بعدها اكتب عمود بعمود																															
	<table><tr><td>C4</td><td>C2</td><td>C5</td><td>C1</td><td>C6</td><td>C3</td></tr><tr><td>C</td><td>O</td><td>M</td><td>P</td><td>U</td><td>T</td></tr><tr><td>E</td><td>R</td><td>S</td><td>S</td><td>E</td><td>C</td></tr><tr><td>U</td><td>R</td><td>I</td><td>T</td><td>Y</td><td>X</td></tr></table>						C4	C2	C5	C1	C6	C3	C	O	M	P	U	T	E	R	S	S	E	C	U	R	I	T	Y	X		
C4	C2	C5	C1	C6	C3																											
C	O	M	P	U	T																											
E	R	S	S	E	C																											
U	R	I	T	Y	X																											
	<table><tr><td>C1</td><td>C2</td><td>C3</td><td>C4</td><td>C5</td><td>C6</td></tr><tr><td>P</td><td>O</td><td>T</td><td>C</td><td>M</td><td>U</td></tr><tr><td>S</td><td>R</td><td>C</td><td>E</td><td>S</td><td>E</td></tr><tr><td>T</td><td>R</td><td>X</td><td>U</td><td>I</td><td>Y</td></tr></table>						C1	C2	C3	C4	C5	C6	P	O	T	C	M	U	S	R	C	E	S	E	T	R	X	U	I	Y		
C1	C2	C3	C4	C5	C6																											
P	O	T	C	M	U																											
S	R	C	E	S	E																											
T	R	X	U	I	Y																											
	Ciphertext: PSTORRTCXCEUMSIUEY																															
	a)	PEYORIUCJERTUQ MST	b)	CERORIMSTPEYUCJT UQ	c)	PEYUCJTUQMSTORI CER	d)	None																								
19.	Validation error , domain error , and boundary condition violation are types of _____																															
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault																								
20.	For exchanging symmetric key among 10 user, we need _____ exchange operation																															
	$n = 10, \frac{n \times n - 1}{2} = \frac{10 \times 9}{2} = 45$																															
	a)	45	b)	100	c)	90	d)	1000																								
21.	_____ is a human mistake in performing some software activity that may lead to a fault in computer program																															
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault																								
22.	Erasure of program or data or malfunction of an OS file manager is considered _____																															
	a)	Modification	b)	Interception	c)	Interruption	d)	Fabrication																								
23.	_____ allow key exchange without real-time access to public-key authority																															
	a)	Certificate authority	b)	Public key authority	c)	Public announcement	d)	Public directory																								
24.	Eavesdropping and impersonate are attacks of _____ Respectively																															
	a)	Availability , Confidentiality	b)	Availability , Integrity	c)	Confidentiality , Integrity	d)	Integrity , Confidentiality																								



25.	In _____, Users interact with directory to obtain any desired public key securely																																
	a)	Certificate authority	b)	Public Key authority	c)	Public announcement	d)	Public Discovery																									
26.	Authentication based on <u>Something the individual has</u> as:																																
	a)	ID	b)	Password	c)	ATM card	d)	PIN																									
27.	_____ is a mistake in <i>interpreting a requirement</i> or a syntax error in a piece of code																																
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault																									
28.	When <u>OS runs the applications</u> of different security requirements <u>under illusion that no other processes exist</u>																																
	a)	Logical separation	b)	Physical separation	c)	Temporal separation	d)	Space separation																									
29.	Consist of replacing the valid source and/or destination IP address with false ones																																
	a)	Smurf attack	b)	Spoofing attack	c)	DoS	d)	Sniffing attack																									
30.	The _____ is a code embedded in program that is set to explode when certain condition is/are met																																
	a)	Trojan house	b)	Trapdoor	c)	Worm	d)	Logical bomb																									
31.	_____ perform simple pattern-matching based on logon and session activity to detect attack																																
	a)	Heuristic -based IDS	b)	Personal Firewall	c)	Proxy	d)	Signature-based IDS																									
32.	Firewall is an executable code runs on a dedicated device to protect inside from outside networks																																
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner																									
33.	The result of encoding a plaintext "TECHNOLOGY" based on playfair cipher with key "SKY" , consider Q is the less character																																
	<table><tr><td>S</td><td>K</td><td>Y</td><td>A</td><td>B</td></tr><tr><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td></tr><tr><td>H</td><td>I</td><td>J</td><td>L</td><td>M</td></tr><tr><td>N</td><td>O</td><td>P/Q</td><td>R</td><td>T</td></tr><tr><td>U</td><td>V</td><td>W</td><td>X</td><td>Z</td></tr></table>				S	K	Y	A	B	C	D	E	F	G	H	I	J	L	M	N	O	P/Q	R	T	U	V	W	X	Z				
	S	K	Y	A	B																												
C	D	E	F	G																													
H	I	J	L	M																													
N	O	P/Q	R	T																													
U	V	W	X	Z																													
a)	PGHNORIREB	b)	PGNUPRIREB	c)	QGNUGRIREB	d)	QGHUOQIREB																										
34.	Consider the cipher text obtained from Q33, Encrypting it with row transposition algorithm , K=2, results _____																																
	هنا عملنا صفين بما ان الـ k يساوي 2 وكتبنا اول حرف فوق ثاني حرف تحت ثالث حرف فوق رابع حرف تحت وهكذا...																																
	<table><tr><td>P</td><td>H</td><td>O</td><td>I</td><td>E</td></tr><tr><td>G</td><td>N</td><td>R</td><td>R</td><td>B</td></tr></table>				P	H	O	I	E	G	N	R	R	B																			
P	H	O	I	E																													
G	N	R	R	B																													
	a)	PNPIEGURRB	b)	PHOIEGNRRB	c)	QNGIEGURRB	d)	QGHUOQIREB																									
35.	Based on the play fair matrix of Q33 , the plaintext of the cipher text "RYTNEW" is _____																																
	a)	Hello	b)	Sunset	c)	Party	d)	None																									



Case 1	A school wants to allow its students to retrieve any information from www. resources on the internet. to help provide efficient service, the school wants to know what sites have been visited and what files from those sites have been fetched; particularly popular files will be cached locally.
Case 2	University students and employees is offices use cable moderns or DSL connections with limited access.

